IISC WORKSHOP · APRIL 2026

Build Your Research Agent

In this session, you build the scaffolding around a language model — context, tools, methodology, and agents — that turns it into something capable of doing real, specific, knowledge-intensive work. You watch it built from scratch, layer by layer. Then you build your own.

PRESENTERS
Rachitt & Vishwajit

VENUE
IISc Bangalore · April 2026

REPO
github.com/v7t-codes/research-agent-workshop

WHY THIS MATTERS TO YOU

The practitioner case

Scale your output

- Research **10x faster** — literature reviews in minutes, not days
- Run **parallel threads** — multiple agents exploring simultaneously
- **Auto-verification** — a second process audits every claim

Apply to real problems

- **Domain-agnostic** — company building, diligence, market analysis
- Any **knowledge-intensive** task becomes agentic
- The harness is **reusable** — swap the domain, keep the system

The compounding edge

- **Cursor, Sierra, Decagon** do this at scale — you're building a narrow version
- The **window is still open** — most practitioners haven't built this yet
- Capability is cheap; **closing the gap to useful output** is the skill

> The harness gets absorbed into the platform over time. **The durable skill is knowing how to close the gap between what a model can do and what produces useful output.**

4:00 – 5:00 PM

First half: we show it

LIVE DEMO — 6 STEPS

0	4:00	**Raw Claude** No setup. Baseline score.
1	4:05	**CLAUDE.md** 20-line context file. Domain, sources, constraints.
2	4:12	**SKILL.md** Research methodology. Decompose, search, synthesize.
3	4:20	**MCP server** 50 lines of Python. Live arXiv and Semantic Scholar.
4	4:30	**Iterative skill** Loop logic. Claude evaluates its own coverage.
5	4:40	**Verification agent** Separate Claude subprocess. Audits every citation.
6	4:50	**Team of agents** Searcher → Critic

What you'll see

- **Live API calls to arXiv** — real 2025 papers fetched in session
- A **second process spawning** — the verification agent runs in its own context
- **Agents writing files** — searcher, critic, synthesizer communicate through disk

The key shift

- **Steps 1–4** run in one session — same context window, same agent
- **Step 5** spawns an independent subprocess — a fresh Claude with no history
- That **independence is why verification works** — no confirmation bias from the original session

5:00 – 7:00 PM

Second half: you build it

GROUPS OF 2–3 · 90 MINUTES

What you're building

- A **research agent for your own domain** — not ours
- Your own CLAUDE.md, SKILL.md, and a custom MCP tool calling a real API
- Agent architecture: verification subprocess or full searcher → critic → synthesizer team

One rule

- You can **read the demo** as reference — you cannot copy it
- Different domain, different methodology, different tool — **write it yourself**

The competition

- At 6:30 PM, your agent runs against **3 unseen research questions**
- **Average RACE score** across all 3 = your result
- Top 3 present: score progression, one design decision, one surprise
- Audience votes separately for **best design** — a creative tool that scores 70 beats a copied one that scores 80

What wins

- **Highest score** — comprehensiveness, insight, verified citations
- **Best design** — the most interesting architectural choice, not the highest number

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